ArcGIS Experience Builder: Customizing and Extending

Christine Wiltawsky & Diego Diaz Doce

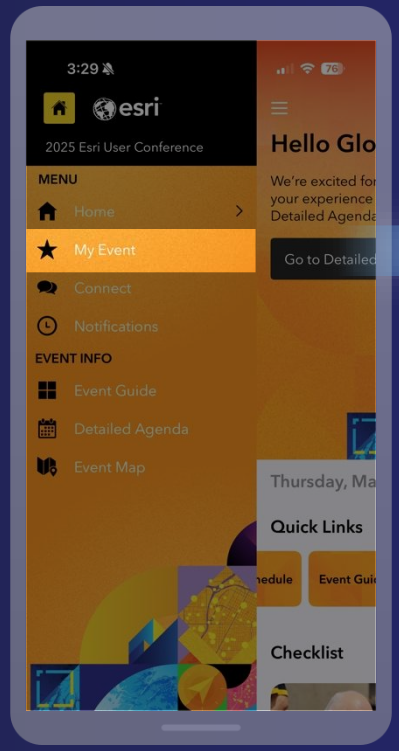


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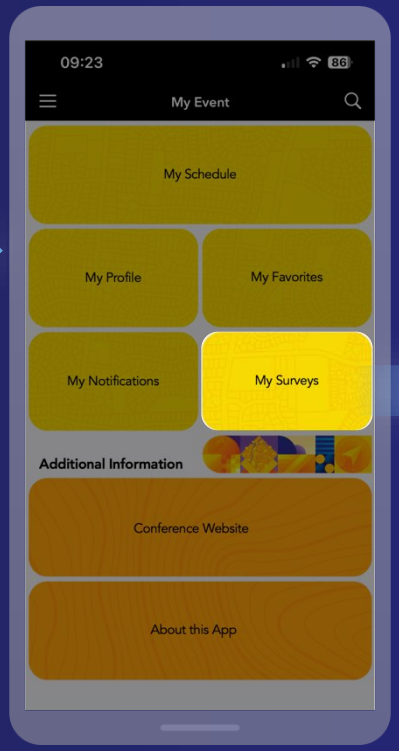
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Please share your feedback in the app

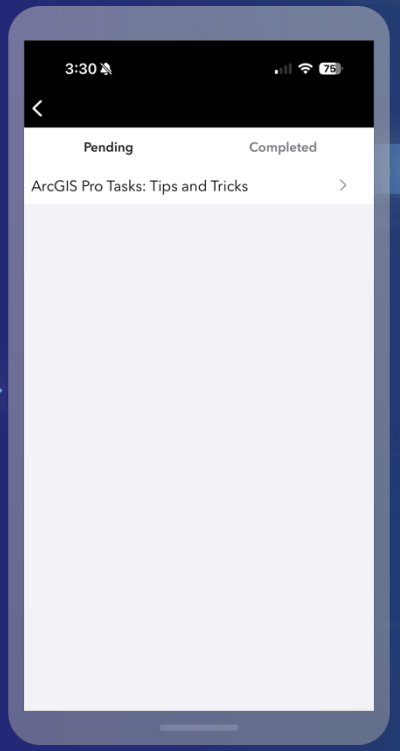
Select “My Event” in the event app main menu



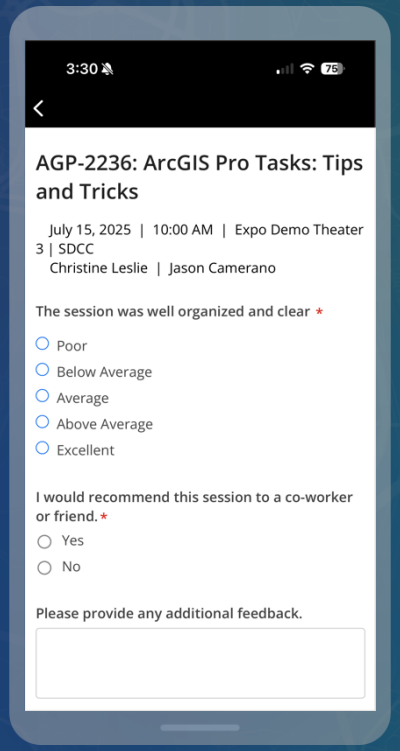
Select “My Surveys”



Select the desired survey



Complete the survey



Agenda

ArcGIS Experience Builder: Customizing and Extending

- Intro to Experience Builder Developer Edition
- Deployment patterns
- Installation
- Custom Widget Development Basics
- Using ArcGIS Maps SDK for JS
- Widget Communication
- AI Capabilities in Experience Builder
- Questions

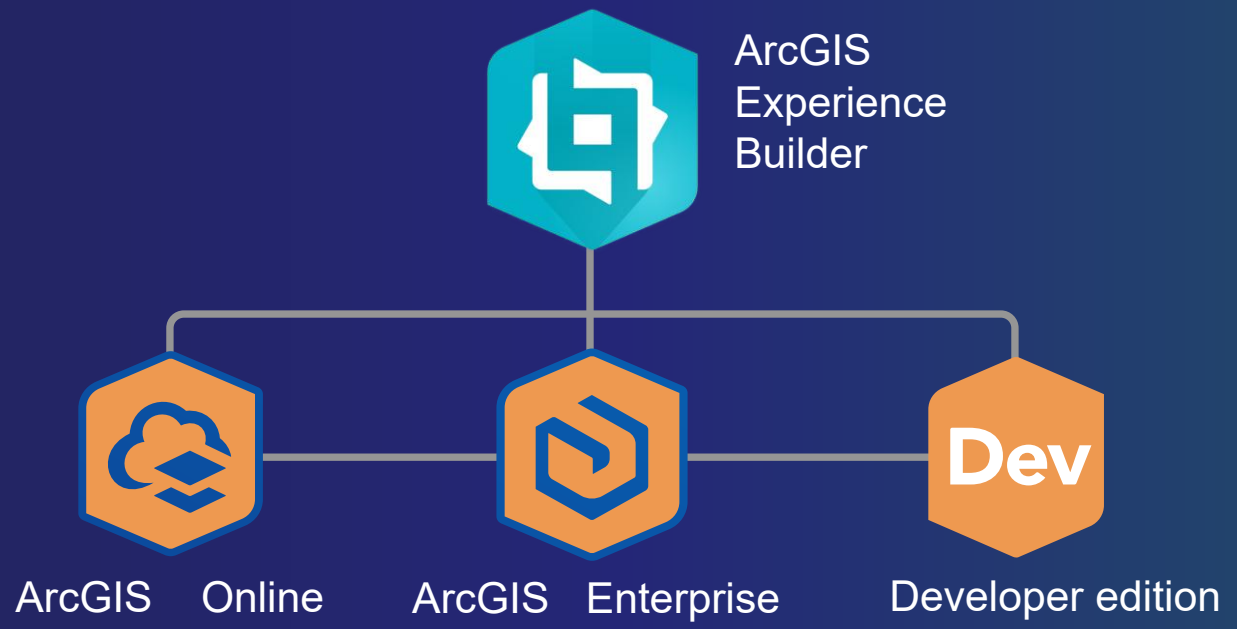
Intro to Experience Builder Developer Edition

Christine Wiltawsky



Experience Builder Editions

Intro to Experience Builder Developer Edition



Key Technologies

HTML, CSS, JavaScript

developer.mozilla.org

TypeScript

typescriptlang.org/docs

React, JSX

react.dev/learn

ArcGIS Maps SDK for JavaScript

esriurl.com/js-tutorials

Deployment patterns

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Widget & App Deployment

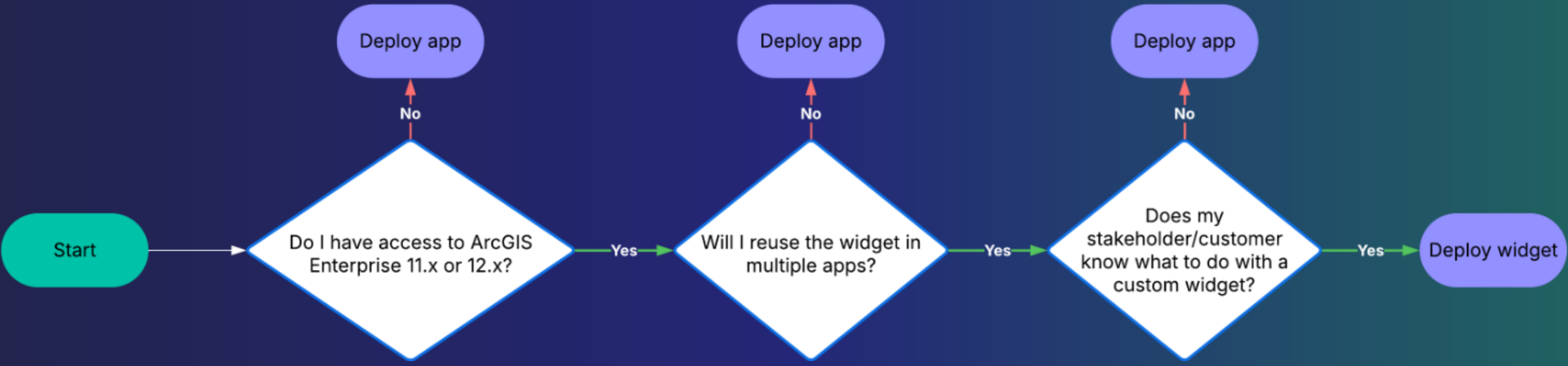
Two patterns for deployment

- Deploy Custom Widget
  - Target is ArcGIS Enterprise
  - Host the custom widget on a web server
  - Register the custom widget in the portal
  - Users within ArcGIS Enterprise can build Experiences with those custom widgets
  - Detailed instructions: esriurl.com/exb-widget-enterprise

- Deploy Custom App
  - Target is a Web Server
  - Build the Experience with your custom widgets
  - Download as ZIP
  - Host the files on a web server
  - Command for automated deployments
  - Detailed instructions: https://developers.arcgis.com/experience-builder/guide/experience-deployment/

Widget & App Deployment

Two patterns for deployment



Installation

Diego Diaz Doce



How to Install ExB Developer Edition

Installation

- Choose a suitable ExB Developer Edition version
- Download and install Node.js
- Download ExB Developer Edition: unzip it to your local drive
- Register an application in either AGOL or Portal
- Install ExB dependencies and start services
- Follow detailed installation steps at:
  - https://developers.arcgis.com/experience-builder/guide/install-guide/

Choosing a suitable ExB Developer Edition version

Installation

- Custom Widget Deployment: version compatible with your ArcGIS Enterprise
- Standalone App Deployment: typically, latest version

exbVersion (Developer edition)	JSAPI version	ArcGIS Enterprise	ArcGIS Online	Calcite version	Recommended Node.js version	Recommended pnpm version	React.js version
1.21	5.1		June 2026	5.1	24	11	19
1.20	5.0	12.1	March 2026	5.0	24		19
1.19	4.34		October 2025	3.3.3	22		19
1.18	4.33	12.0	June 2025	3.2.1	22		18
1.17	4.32	11.5	March 2025	3.0.3	22		18

https://developers.arcgis.com/experience-builder/guide/release-versions/

Custom Widget Development Basics

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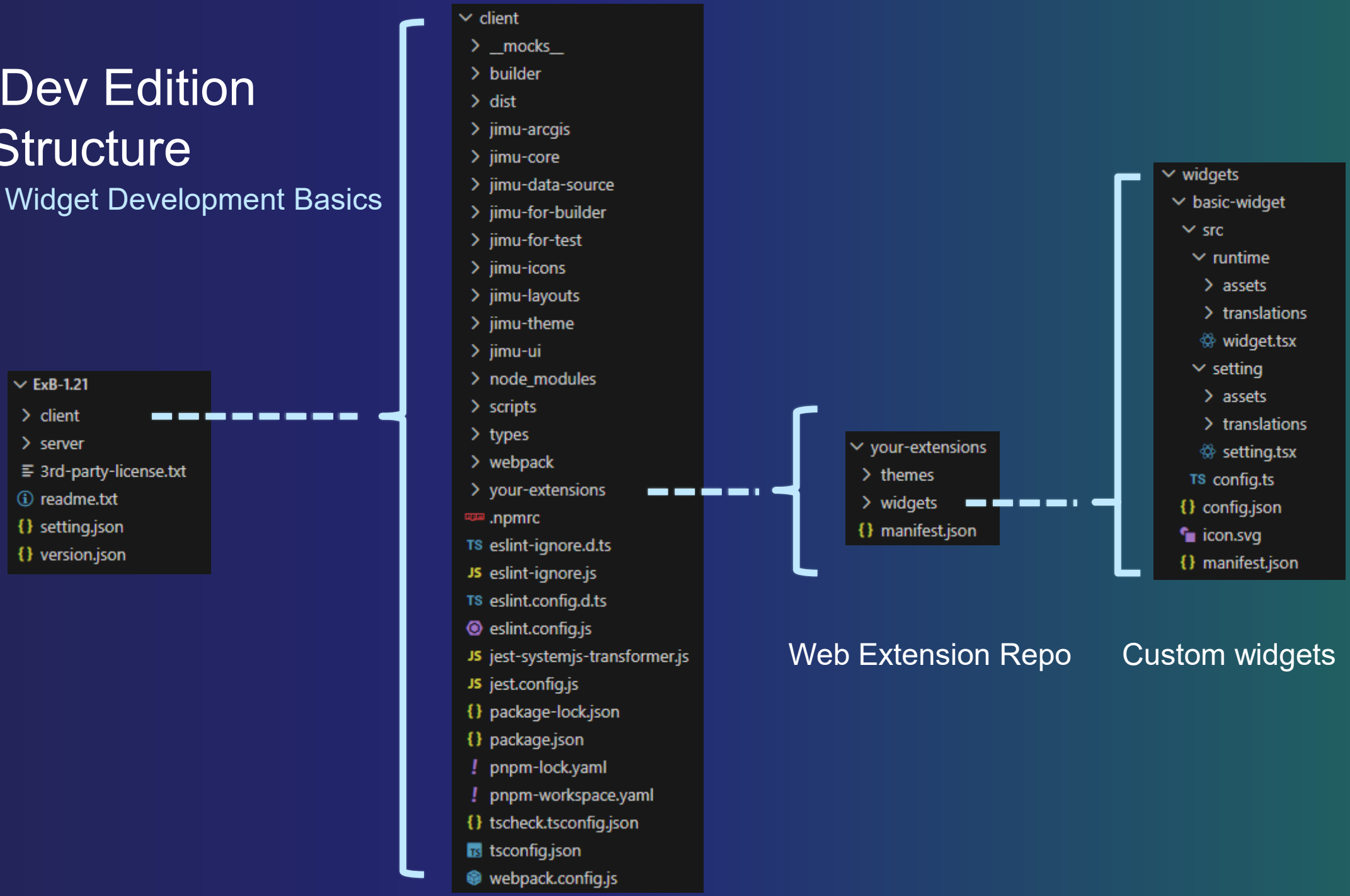
Widgets and Components

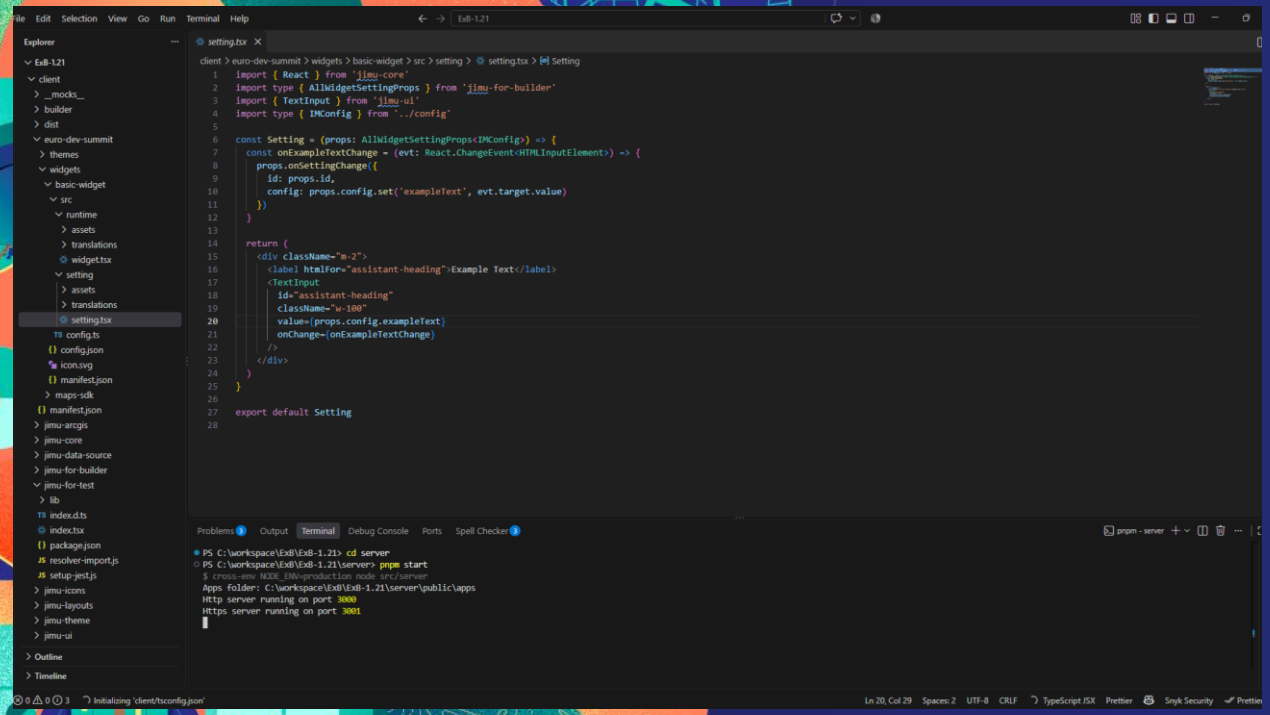
Custom Widget Development Basics

- ExB Widget
  - Building block for your Experience
  - Can be added from the ExB widget panel
- React Component
  - Self-contained UI element
  - Used inside custom widgets
- Maps SDK Widget
  - Traditional Maps SDK UI control
  - Now deprecated
  - Example: new Search({view});
- Maps SDK Web Component
  - Modern Web Component version of the Maps SDK Widget
  - Example: <arcgis-search></arcgis-search>

ExB Dev Edition File Structure

Custom Widget Development Basics





Demo: Basic Widget

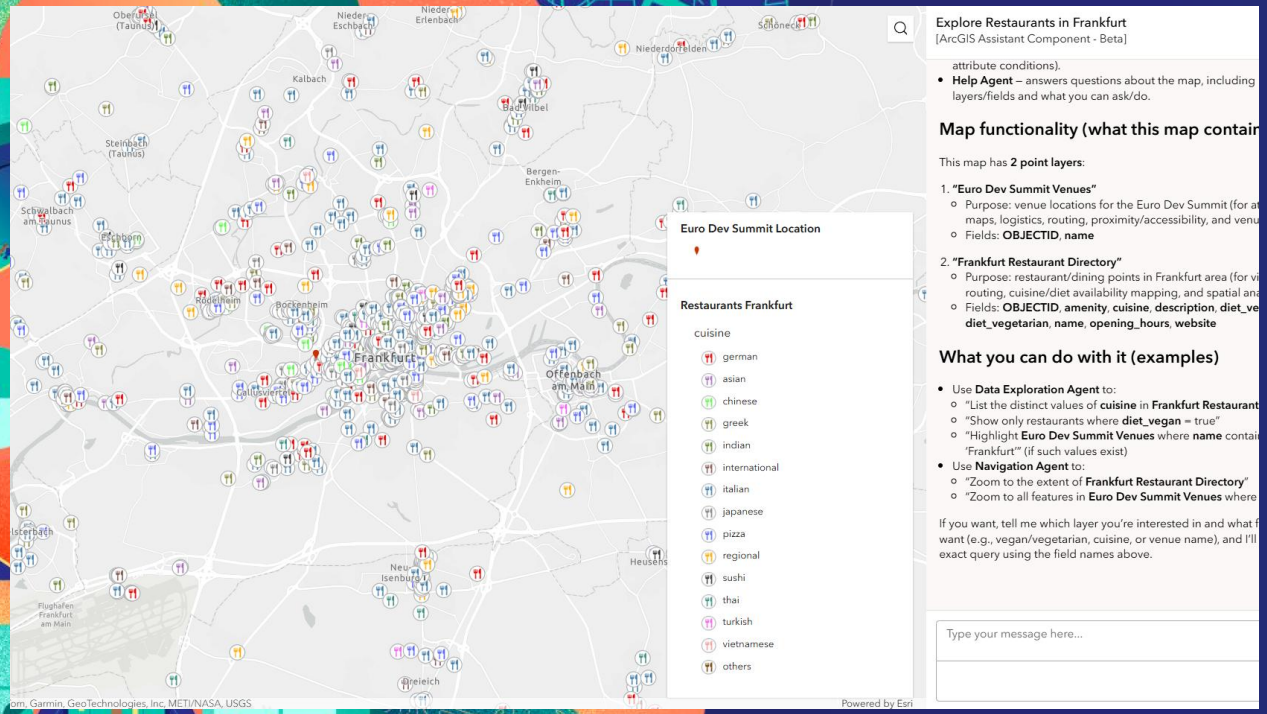
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- Experience with custom widget
- Settings panel
- Using config

Using ArcGIS Maps SDK for JS

Diego Diaz Doce





Demo: Using ArcGIS Maps SDK for JS

Diego Diaz Doce

- Enabling the ArcGIS Maps SDK for JS
- Accessing the map: JimuMapViewComponent
- Adding a legend component inside the map widget
- Adding an AI component inside the custom widget

Migrating ExB Custom Widgets to Web Components

Using ArcGIS Maps SDK for JS

- Your Custom Widgets should use Maps SDK Web Components, not Maps SDK Widgets
- ArcGIS Maps SDK for JavaScript widgets retirement timeline
  - Deprecation: Q1 2026 | Retirement : Q1 2027
- What should I do with my custom widgets?
  - No action needed if not using Maps SDK widgets
  - New custom widgets: use Maps SDK Web Components
  - Existing custom widgets: migrate them to Maps SDK Web Components ASAP
- Custom widgets, using Maps SDK widgets, will stop working when:
  - You upgrade to an ExB Developer Edition version released in Q1 2027 or later
  - You add custom widgets to ArcGIS Enterprise released in the first half of 2027 or later.

Widget Communication

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DataSource

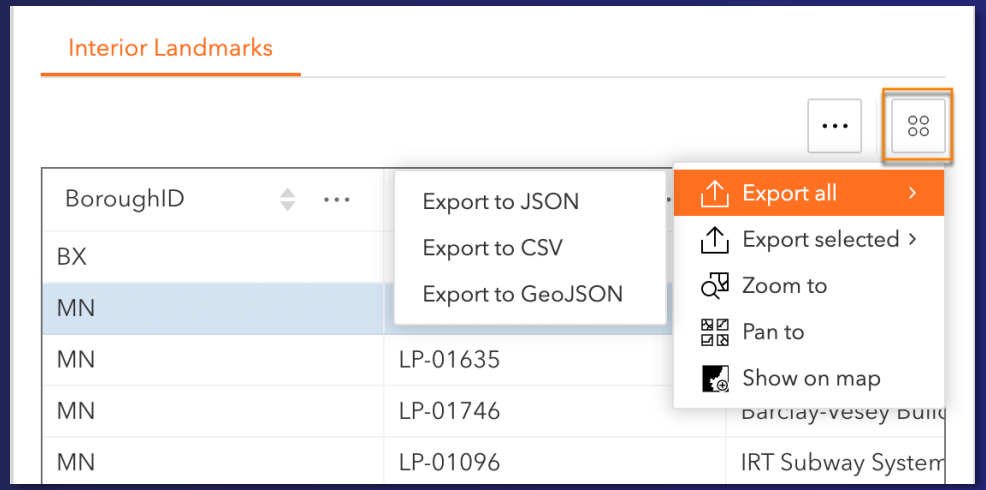
Widget Communication

- Selection with DataSourceManager
- QueryParameter (filter) in DataSource
- Output DataSource

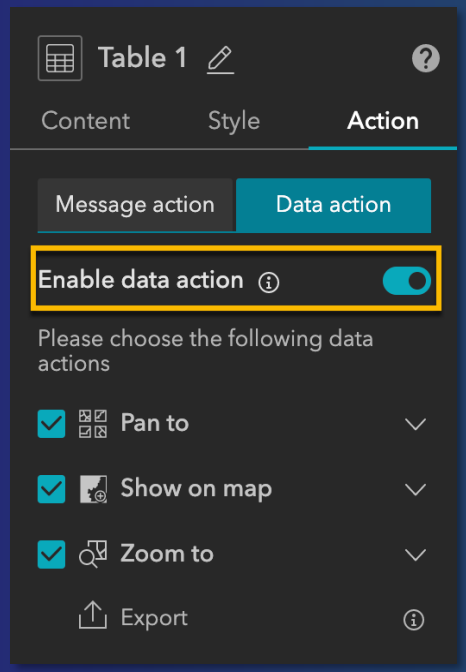
Actions

Widget Communication

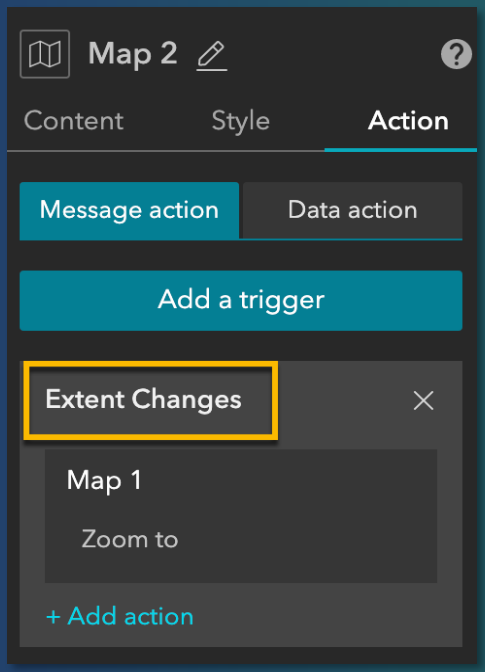
- Widgets communicate with each other
- Data actions are opt-in
- Message actions are automatic



Data action



Message action



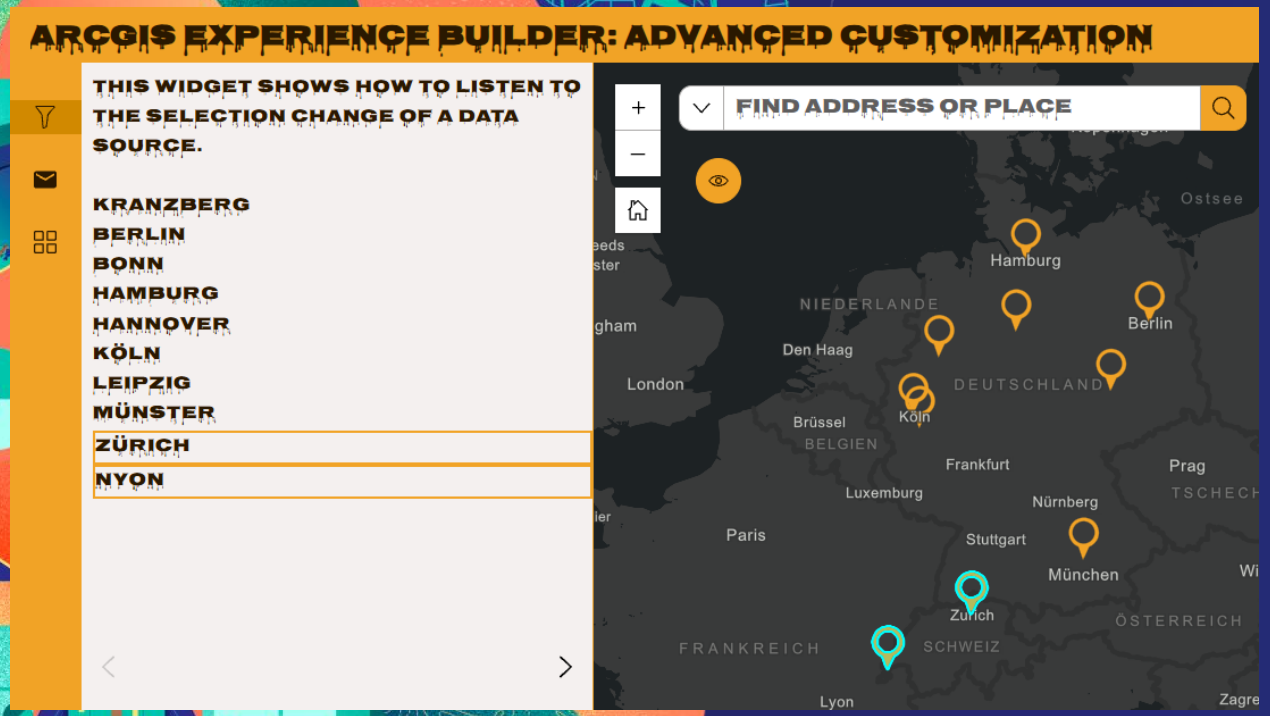
Widget State in Redux Store

Widget Communication

- Use the AppStore to monitor other widgets' states

- Add / Update custom properties in other widgets' states
  - AppActions and AppStore for simple objects
  - MutableStoreManager and mutableStateProps for complex objects

- Use App Store and AppActions to open or close other widgets



Demo: Widget Communication

Christine Wiltawsky

Widget Lifecycle

Mirror, mirror on the wall, am I visible at all?



Demo: Widget Lifecycle

Christine Wiltawsky

AI Capabilities in Experience Builder

Christine Wiltawsky & Diego Diaz Doce



AI Capabilities in Experience Builder

Q4 2026

AI Chat for End Users (Beta)

End users can interact with AI through a chat window to explore content, ask questions, and perform supported actions within the app

Q1 2027

Builder Assistant for App Authors (Beta)

Help app authors create apps easier & more intuitive

Available now

AI Coding Skill for Widget Developers (Open source)

Provides AI coding tools with Experience Builder context and best practices to develop production-ready custom widgets.

Make Your Widget AI Ready

- Runtime (AI Chat)
  - Provide AI actions, AI renderers, and human inputs for human-in-the-loop (HITL) workflows.
- Setting (Builder assistant)
  - Provide config AI actions so AI can configure widget settings.

Experience Builder developer guide
👉Add AI to Your Widget (v1.22, December 3)

AI Agent Skill for ExB Widget Development

- What is it?
  - Set of instructions that provide coding agents with guidance for building ExB custom widgets
  - Works with any skills-compatible agent
- Why should I use it?
  - Accelerates widget development
  - Promotes consistency and best practices
  - Output quality = Agent Skill x LLM x Prompt
- How can I start using it?
  - Get the skill: https://github.com/esri/arcgis-experience-builder-sdk-resources
  - Follow this tutorial: https://www.esri.com/arcgis-blog/products/experience-builder/geoai/building-custom-experience-builder-widgets-with-an-agent-skill

Resources

https://github.com/EsriDevEvents/arcgis-experience-builder-advanced-customization-dtseu-2026

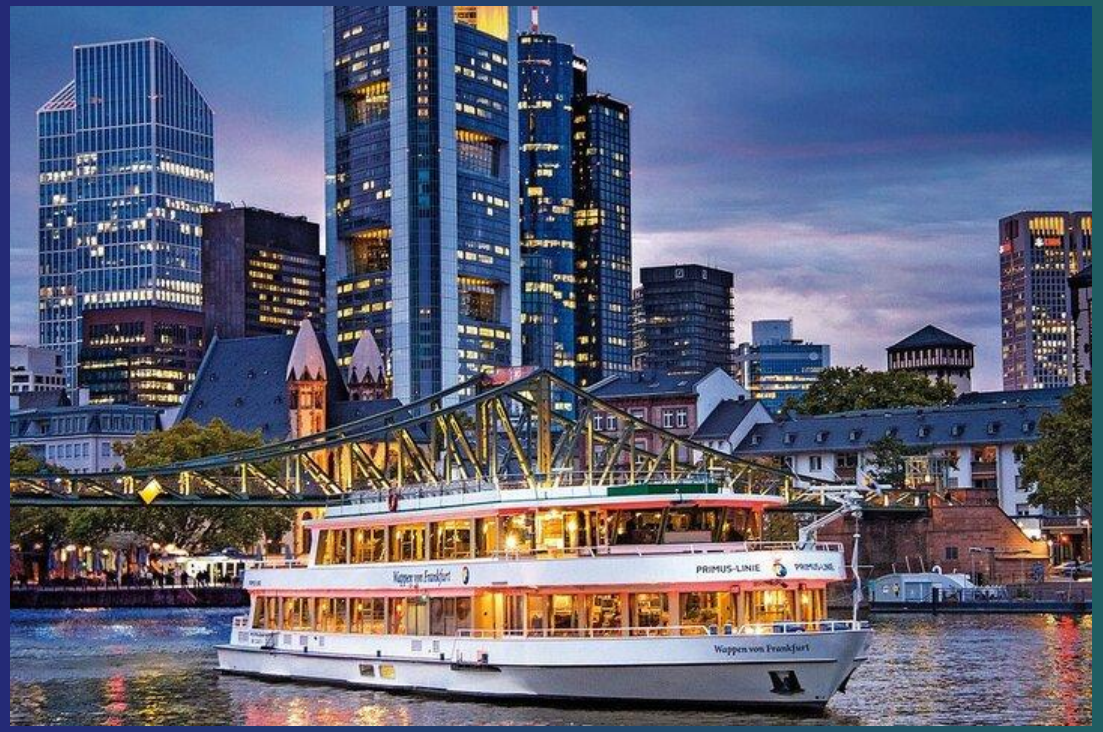
Questions



Join us for tonight’s Dev & Tech Summit Party!

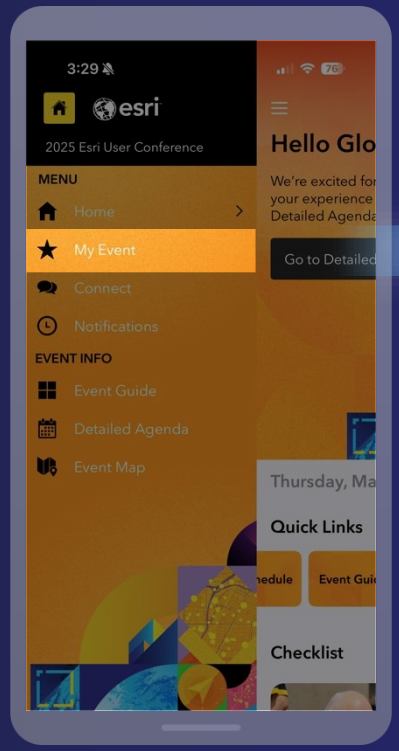
Mainkai 36 60311 Frankfurt am Main

- Collect your ticket with boat assignment from the Registration Desk from 10:00 – 16:30.
- Shuttles will begin loading after sessions conclude at 17:00.
- Boats will depart promptly at **18:15**, be sure to arrive early!

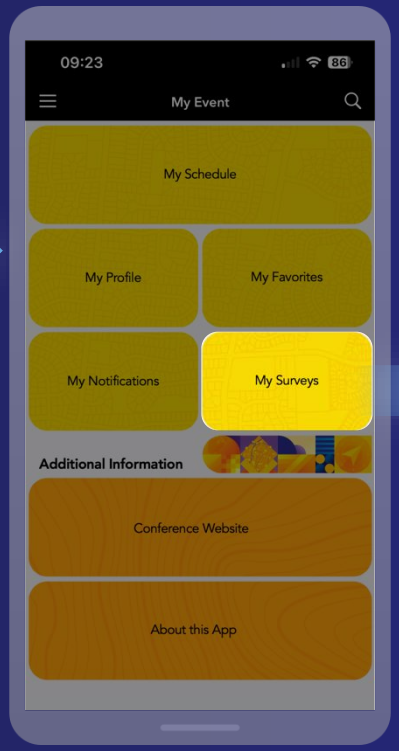


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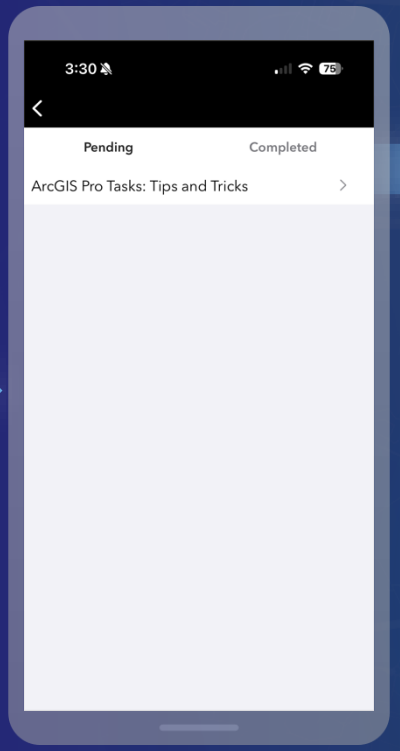
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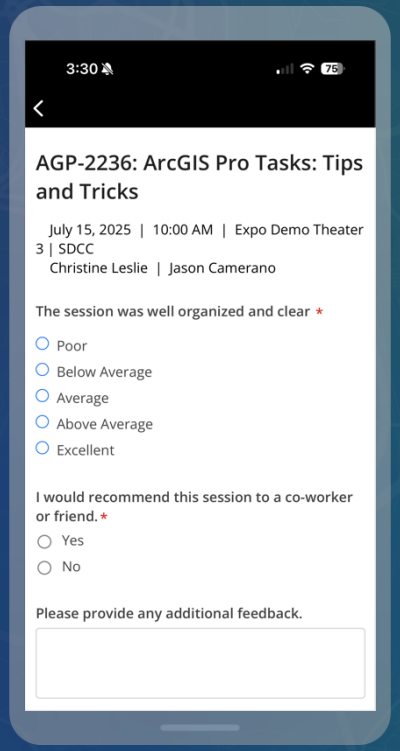
Select “My Surveys”



Select the desired survey



Complete the survey



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